

Cordyceps militaris hot water extract inhibits lipopolysaccharide-induced inflammatory response in porcine alveolar macrophages by the regulation of mitogen-activated protein kinase signaling pathway

Felix Shih-Hsiang Hsiao, Yeong-Hsiang Cheng, Szu-Kai Wang, and Yu-Hsiang Yu

Abstract: *Cordyceps militaris* is a rare and exotic medicinal mushroom used in traditional Chinese medicine. The secondary metabolite, cordycepin (3'-deoxyadenosine), produced from *C. militaris* is a biologically active compound. Cordycepin has been demonstrated to exert several pharmacological effects, such as anti-microbial and anti-tumor activities. However, the effect of cordycepin on the immune modulation of porcine alveolar macrophages (PAM) is poorly investigated. In the current study, we investigated the immunomodulatory effect of cordycepin from *C. militaris* hot water (CMHW) extract on lipopolysaccharide (LPS) simulated PAM. The CMHW significantly reduced LPS-stimulated nitric oxide production and cyclooxygenase-2 protein expression levels in a dose-dependent manner. The LPS-induced phosphorylation of p38 was impaired by CMHW treatment, thereby decreasing pro-inflammatory cytokine secretion, including tumor necrosis factor- α , interleukin-1 β , and interleukin-6. Similar to CMHW, cordycepin also had similar effects on LPS-stimulated PAM. Taken together, these findings demonstrate that CMHW has an inhibitory effect on LPS-induced inflammatory responses in PAM through regulation of the p38 mitogen-activated protein kinase signaling pathway. Thus, CMHW is a potential novel feed additive for immunomodulation in farm animals.

Key words: *Cordyceps militaris*, cordycepin, porcine alveolar macrophage, lipopolysaccharide, immunomodulation.

Résumé : *Cordyceps militaris* est un champignon médicinal rare et exotique utilisé dans la médecine traditionnelle chinoise. Un métabolite secondaire, la cordycépine (3'-désoxyadénosine), produit par *C. militaris*, est un composé biologiquement actif. Il a été démontré que la cordycépine exerce plusieurs effets pharmacologiques, comme des activités anti-microbiennes et anti-tumorales. Par contre, l'effet de la cordycépine sur la modulation immunologique des macrophages alvéolaires de porc (PAM – « porcine alveolar macrophages ») n'a été que très peu étudié. Dans la présente étude, nous avons examiné l'effet immunomodulateur de la cordycépine provenant d'un extrait à l'eau chaude de *C. militaris* (CMHW – « *C. militaris* hot water ») sur les PAM stimulés par lipopolysaccharide (LPS). Le CMHW a diminué de façon significative la production d'acide nitrique stimulé par le LPS et les niveaux d'expression de la protéine cyclo-oxygénase-2 de manière proportionnelle à la dose. La phosphorylation induite par LPS de p38 a été altérée par le traitement CMHW, réduisant ainsi la sécrétion de cytokines pro-inflammatoires, incluant le facteur de nécrose des tumeurs α , interleukine-1 β et interleukine-6. Semblable au CMHW, la cordycépine avait des effets similaires sur les PAMs stimulés par LPS. Dans leurs ensembles, ces résultats démontrent que le CMHW a un effet inhibiteur sur les réponses inflammatoires induites par LPS dans les PAM par régulation de la voie de signalisation de la protéine kinase activée par mitogène p38. Donc, le CMHW est une source potentielle d'un nouveau supplément alimentaire pour l'immunomodulation chez les animaux de ferme. [Traduit par la Rédaction]

Mots-clés : *Cordyceps militaris*, cordycépine, macrophage alvéolaire porcin, lipopolysaccharide, immunomodulation.

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F.S.H. Hsiao,* Department of Animal Science and Biotechnology, Tunghai University, Taichung 407, Taiwan.

Y.-H. Cheng,* S.-K. Wang, and Y.-H. Yu. Department of Biotechnology and Animal Science, National Ilan University, Yilan 260, Taiwan.

Corresponding author: Yu-Hsiang Yu (email: yuyh@niu.edu.tw).

*These authors contributed equally to this work.

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Introduction

Cordyceps militaris is a rare and exotic Chinese medicinal mushroom. Its chemical composition and pharmacological effects are similar to *Cordyceps sinensis*. *Cordyceps sinensis* is widely used as a tonic for vitality and longevity for thousands of years by the Chinese (Zhu et al. 1998). *Cordyceps militaris* has also been used in Chinese medicine for thousands of years. Medicinal fungi have been traditionally used to treat hyposexuality, hyperlipidemia, asthma, and lung inflammation (Huang et al. 2004; Wang et al. 2007; Guo et al. 2010). Recently, it has been reported that *C. militaris* exerts several pharmacological effects, such as anti-oxidant, anti-inflammatory, anti-microbial, anti-tumor and anti-angiogenic activities, and immunopotential (Ng and Wang 2005; Kuo et al. 2007).

The medical potential of the metabolites of *C. militaris* has been widely reviewed (Das et al. 2010b; Tuli et al. 2013). Cordycepin, or 3'-deoxyadenosine, is the major active secondary metabolite of *C. militaris*. The chemical structure of cordycepin is similar to adenosine. It has been reported that the nucleoside analogue exhibits anti-microbial, anti-fungal, and anti-tumor activities (Sugar and McCaffrey 1998; Ahn et al. 2000; Nakamura et al. 2006). Cordycepin also has anti-inflammatory effects in lipopolysaccharide (LPS) induced murine macrophages and microglia (H.G. Kim et al. 2006; Jeong et al. 2010).

Several methodologies have been developed to improve cordycepin productivity, such as liquid culture system or extraction methods (Das et al. 2008, 2010a; Ni et al. 2009; Wang et al. 2014). However, these methodologies still require high capital investments and operating costs. Alternatively, it has been demonstrated that cordycepin can be produced by solid-state fermentation (Cheng et al. 2016) and subsequent extraction from *C. militaris* by water extraction (G.Y. Kim et al. 2006; Jo et al. 2010). These methodologies are feasible and relatively low cost. However, the extraction procedure and bioactivity of cordycepin still remain to be optimized. In the past decade, studies about the pharmacological and biochemical mechanisms of *C. militaris* or cordycepin have all focused on human or mouse models. Recently, we demonstrated that feed supplemented with *C. militaris* fermentation products improves growth performance and enhances cell-mediated immunity in piglets (Cheng et al. 2016). Porcine alveolar macrophages (PAM) are the naturally permissive host cells for respiratory virus infection, such as porcine reproductive and respiratory syndrome virus. But the biochemical mechanism of the immunomodulatory effect of *C. militaris* hot water (CMHW) on PAM has not been elucidated.

The purpose of this study was to investigate the inhibitory effects of cordycepin from CMHW and the mechanism by which it modulates the anti-inflammation mechanism in LPS-stimulated PAM. These results might

provide valuable information about the effect of CMHW on immunomodulation by PAM for development as a feed additive.

Materials and Methods

Cordyceps militaris culture

The *C. militaris* strain (BCRC 32219) was purchased from the Biosource Collection and Research Center (BCRC, Hsinchu, Taiwan) and was used in solid-state fermentation. *Cordyceps militaris* was transferred to potato dextrose agar (Difco Laboratories, Detroit, MI, USA) and incubated at 22 °C for 7 d. *Cordyceps militaris* was then inoculated into an Erlenmeyer flask containing potato dextrose broth (Difco Laboratories, Detroit, MI, USA) at 22 °C for 5 d with shaking.

Solid-state fermentation of *C. militaris*

For solid-state fermentation, the wheat-based substrates containing 0.1% CaCO₃, 0.05% MgSO₄, 0.1% NaH₂PO₄, 50% H₂O, 0.05% KH₂PO₄, and 1% glucose were inoculated with a level of 10% (v/w) inoculum, mixed carefully under sterile conditions, and were incubated in a chamber with relative humidity above 80%. Cultures were incubated in darkness at 22 °C for 2 wk.

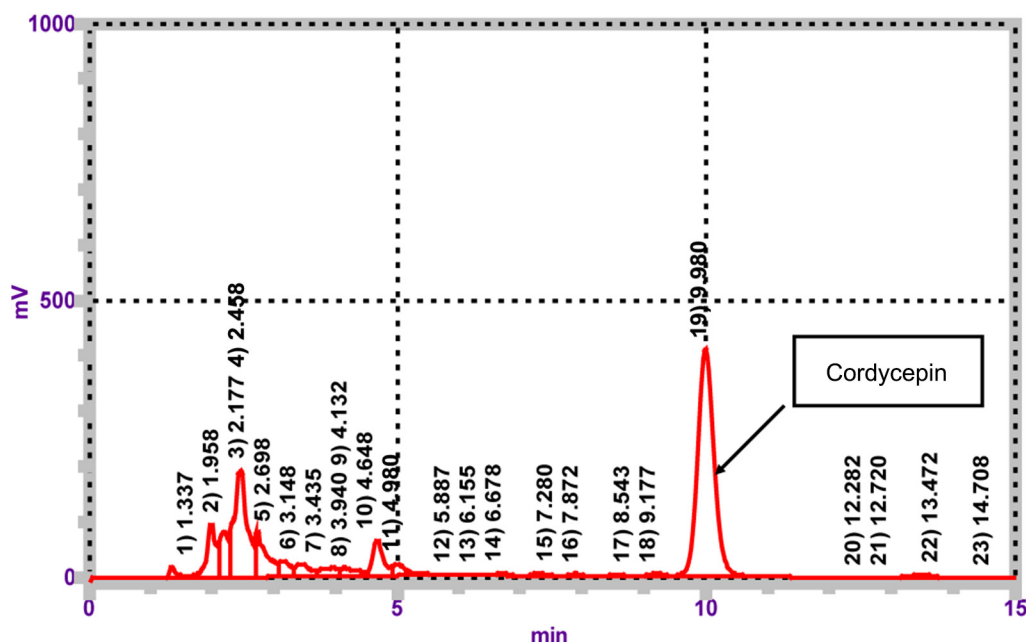
Extraction and analysis of cordycepin

The fermentation product was baked at 50 °C for 24 h and then homogenized by mechanical agitation. The dried fermentation product was heated by distilled deionized water at 100 °C for 1 h and then filtered by a Whatman No. 2 filter. The CMHW filtrate was lyophilized and reconstituted with RPMI-1640 medium (Invitrogen, Carlsbad, CA, USA). The CMHW was then tested for cordycepin content. Cordycepin was quantified according to a previous study with modifications (Chang et al. 2005). In brief, a SPD-10A system (Shimadzu, Columbia, MD, USA) with a programmable UV detector (10A VP, Shimadzu, Columbia, MD, USA) was used in this study. A reverse phase RP-18 column (LiChrospher 100 RP-18 end-capped, 5 µm) protected with a guard was used throughout the experiments. Samples (20 µL) were injected into the high-performance liquid chromatography (HPLC) column. The mobile phase consisted of methanol: 0.02 mol L⁻¹ KH₂PO₄ (15:85, v/v). The flow rate was 1 mL min⁻¹. Cordycepin was determined at a wavelength of 254 nm by use of a UV detector. The recorder was set to 15 min. For cordycepin quantification, linear response was obtained over a range of 100–500 µg mL⁻¹ of the cordycepin standard. Standards were prepared and analyzed a minimum of three times. The concentration of cordycepin was determined based on the slope of the standard curve.

Isolation and culture of PAM

The PAM from 4-wk-old male cross-bred (Yorkshire × Landrace) weanling piglets were isolated by bronchoalveolar lavages using cold phosphate-buffered saline (PBS).

Fig. 1. Identification of cordycepin from *Cordyceps militaris* hot water (CMHW). Measurement of CMHW by high-performance liquid chromatography (mV). Three experiments were carried out, and one representative result is shown. Colour online.



Bronchoalveolar fluids were centrifuged at 150g for 5 min at 4 °C and cell viability was assessed by 0.1% trypan blue exclusion assay. Differential cell counts were performed on cytospin preparations stained with Diff-Quik (Diagnostic Systems, Gibbstown, NJ, USA). The PAM were then frozen in liquid nitrogen in 90% fetal bovine serum (FBS) and 10% dimethyl sulfoxide (DMSO). After thawing, PAM were adjusted to a cell concentration of 2×10^6 cells per mL in growth medium (RPMI-1640 medium containing 10% FBS and 1% penicillin-streptomycin) and cultured 37 °C in 5% CO₂. The PAM were pretreated with the indicated concentrations of the CMHW for 30 min before $1 \mu\text{g mL}^{-1}$ of LPS treatment. After 24 h of LPS stimulation, the supernatant of culture medium from PAM was harvested and cytokine concentrations were determined. Total protein lysate from LPS-stimulated PAM was extracted and the protein content was analyzed by Western blot.

3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assay

The PAM were seeded into 96-well plates at a density of 5×10^3 per well and treated with the indicated concentrations of CMHW for 24 h. For the MTT assay, medium was removed and 20 μL of MTT solution (5 mg mL⁻¹ in PBS) was added into each well. Cells were incubated at 37 °C in 5% CO₂ for 4 h. The medium was then removed and 100 μL of DMSO was added into each well. The plate was gently rotated on an orbital shaker for 10 min. The absorbance was detected at 570 nm with a microplate reader (VersaMax, Molecular Devices, Sunnyvale, CA, USA).

Measurement of nitric oxide (NO)

The NO production was determined by measuring nitrite accumulation. PAM were seeded into 96-well plates at a density of 2×10^5 per well and pretreated with the indicated concentrations of CMHW or cordycepin for 1 h before $1 \mu\text{g mL}^{-1}$ of LPS treatment. After 24 h of LPS stimulation, 50 μL of supernatant was mixed with 50 μL of Griess reagent (0.1% naphthylethylenediamine dihydrochloride, 1% sulfanilamide in 5% H₃PO₄, v/v). After 5 min incubation at room temperature, the optical density of the samples was measured by spectrophotometry at 540 nm.

Western blot

Total protein from PAM was extracted by radioimmunoprecipitation assay buffer with protease and phosphatase inhibitors. The sample was centrifuged at 12 000 rev min⁻¹ for 10 min and supernatant was subjected to Western blot. Protein lysate (20 μg) was separated by sodium dodecyl sulfate-polyacrylamide gel electrophoresis and then transblotted onto a polyvinylidene fluoride membrane (Perkin Elmer, Norwalk, CT, USA). The cyclooxygenase-2 (COX-2) primary antibody was purchased from EMD Millipore (Danvers, MA, USA). The phospho-c-Jun N-terminal kinases (p-JNK), phospho-extracellular signal-regulated kinases (p-ERK), and p-p38 primary antibodies were purchased from Sigma-Aldrich (St. Louis, MO, USA). The JNK, ERK, and p38 primary antibodies were purchased from Santa Cruz Biotechnology (Dallas, TX, USA). β -actin antibody (EMD Millipore, Danvers, MA, USA) was used for the loading control in the lysates of total protein. The secondary

antibody coupled to horseradish peroxidase was used in the chemiluminescence procedure (Immobilon Western, EMD Millipore, Danvers, MA, USA). The Western blotting procedure was performed according to the manufacturer's instruction.

Cytokine assay

The commercial ELISA kits (R&D Systems, Minneapolis, MN, USA) were used to evaluate the accuracy of the interleukin-6 (IL-6), tumor necrosis factor- α (TNF- α), and interleukin-1 β (IL-1 β) contents. The supernatant from the cultured cells was collected and centrifuged at 1500 rev min⁻¹ for 10 min. The optical density in each kit well was detected using a microplate reader (VersaMax, Molecular Devices, Sunnyvale, CA, USA) and the cytokine levels were deduced from the absorbance value by extrapolation from a standard curve generated in parallel.

Ethics statement

All experiments were performed in accordance with the approved guidelines. The animal protocol was approved by the National Ilan University Institutional Animal Care and Use Committee (IACUC Approval No. 104-14).

Statistical analysis

The results were expressed as the percentage subjected to logarithmic transformation prior to analysis of variance. Statistical significance among groups was determined by one-way analysis of variance. Duncan's new multiple range test was used to evaluate differences among means (SAS Institute Inc., Cary, NC, USA). *P* values less than 0.05 were considered statistically significant.

Results

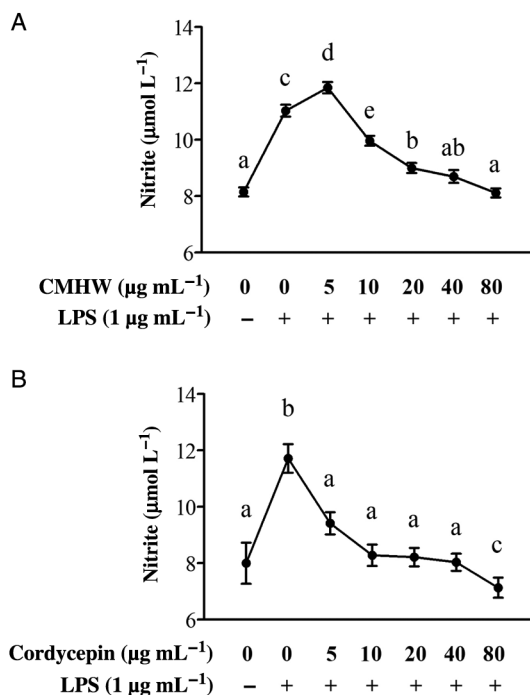
Identification of cordycepin from CMHW

To examine the concentration of cordycepin after hot water extraction, CMHW was analyzed by HPLC based on a standard curve. We found that a specific peak was detected with a chromatograph identical to standard cordycepin (Fig. 1), indicating that CMHW contains a high amount of cordycepin after hot water extraction. To determine the cytotoxic effect of CMHW, PAM were treated with CMHW at various concentrations for 24 h. No significant difference was found in cell survival rate even at the highest concentration of 80 $\mu\text{g mL}^{-1}$ CMHW (data not shown). These findings demonstrate that hot water extraction is a practically feasible method for cordycepin extraction from solid-state fermentation product of *C. militaris*.

Effect of CMHW and cordycepin on inflammatory response in LPS-treated PAM

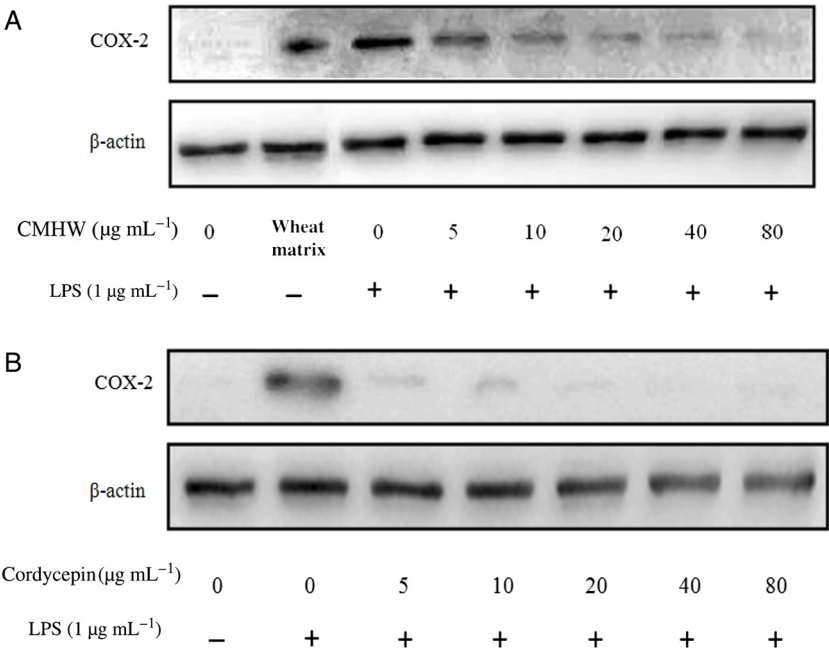
We first investigated whether CMHW could alleviate LPS-induced nitrate biosynthesis in PAM. As expected, LPS significantly promoted nitrate levels in PAM

Fig. 2. Effect of *Cordyceps militaris* hot water (CMHW) and cordycepin on nitric oxide (NO) levels in lipopolysaccharide (LPS) stimulated porcine alveolar macrophages (PAM). The PAM were pretreated with (A) various concentrations of CMHW (0–80 $\mu\text{g mL}^{-1}$) and (B) cordycepin (0–80 $\mu\text{g mL}^{-1}$) for 1 h and then stimulated with 1 $\mu\text{g mL}^{-1}$ LPS for 24 h. The NO from supernatant of culture medium was analyzed using Griess reagent. Values were expressed as mean $\pm$ SEM (*n* = 6). Means with different letters are significantly different (*P* < 0.05).



(Figs. 2A, 2B). However, CMHW obviously attenuated LPS-stimulated nitrate production in a dose-dependent manner (Fig. 2A). Cordycepin also consistently had an inhibitory effect on nitrate production (Fig. 2B). We then investigated the effects of CMHW and cordycepin on COX-2 protein expression in LPS-stimulated PAM by Western blot analysis. Interestingly, fermentation substrate, the wheat matrix, significantly increased the protein level of COX-2 in PAM (Fig. 3A), indicating that wheat matrix contains potent pro-inflammatory factors. The LPS treatment further promoted COX-2 protein expression in PAM, whereas CMHW significantly alleviated LPS-induced COX-2 protein expression in a concentration-dependent manner at treatments up to 80 $\mu\text{g mL}^{-1}$ (Fig. 3A). Similarly, cordycepin also suppressed COX-2 protein expression in LPS-stimulated PAM (Fig. 3B). Similarly, wheat matrix significantly increased the pro-inflammatory cytokine levels in the culture media, such as TNF- α and IL-1 β (Fig. 4A). By contrast, we found no significant effects of wheat matrix on IL-6 levels (Fig. 4A). Furthermore, pro-inflammatory cytokines were increased in the culture media of LPS-stimulated PAM (Figs. 4A–4C), whereas CMHW remarkably attenuated LPS-stimulated

Fig. 3. Effect of *Cordyceps militaris* hot water (CMHW) and cordycepin on lipopolysaccharide (LPS) induced cyclooxygenase-2 (COX-2) protein expression in porcine alveolar macrophages (PAM). The PAM were pretreated with (A) various concentrations of CMHW (0–80 $\mu\text{g mL}^{-1}$) and (B) cordycepin (0–80 $\mu\text{g mL}^{-1}$) for 1 h and then stimulated with 1 $\mu\text{g mL}^{-1}$ LPS for 24 h. The protein levels of COX-2 and β -actin were determined by immunoblotting analysis. Three experiments were carried out, and one representative result is shown.



pro-inflammatory cytokine production (Figs. 4A–4C). Cordycepin also exerted a similar effect on inhibition of pro-inflammatory cytokine levels in LPS-stimulated PAM (Figs. 4D–4F). These results demonstrate that CMHW and cordycepin showed a significant inhibitory effect on the production of pro-inflammatory mediators and cytokines in LPS-stimulated PAM.

Effect of CMHW and cordycepin on mitogen-activated protein kinase (MAPK) signaling pathway in LPS-treated PAM

We next investigated whether CMHW and cordycepin inhibit LPS-induced pro-inflammatory mediators and cytokine production by regulation of the MAPK pathway. Similarly, phosphorylation of ERK and p38 were elevated by wheat matrix treatment in PAM (Fig. 5A). As expected, LPS significantly induced the phosphorylation of ERK and p38 in PAM (Fig. 5A). The CMHW alleviated the phosphorylation of ERK in LPS-stimulated PAM at low concentration, while this effect was not observed at high concentration of CMHW (Fig. 5A). Interestingly, LPS in combination with CMHW treatment activated the phosphorylation of JNK in PAM compared with LPS treatment alone (Fig. 5A). In addition, CMHW reduced the phosphorylation of p38 in LPS-stimulated PAM in a dose-dependent manner (Fig. 5A). Cordycepin remarkably suppressed the phosphorylation of ERK and p38 in LPS-stimulated PAM in a concentration-dependent manner (Fig. 5B). In contrast, cordycepin attenuated the phosphorylation of JNK in LPS-stimulated PAM in

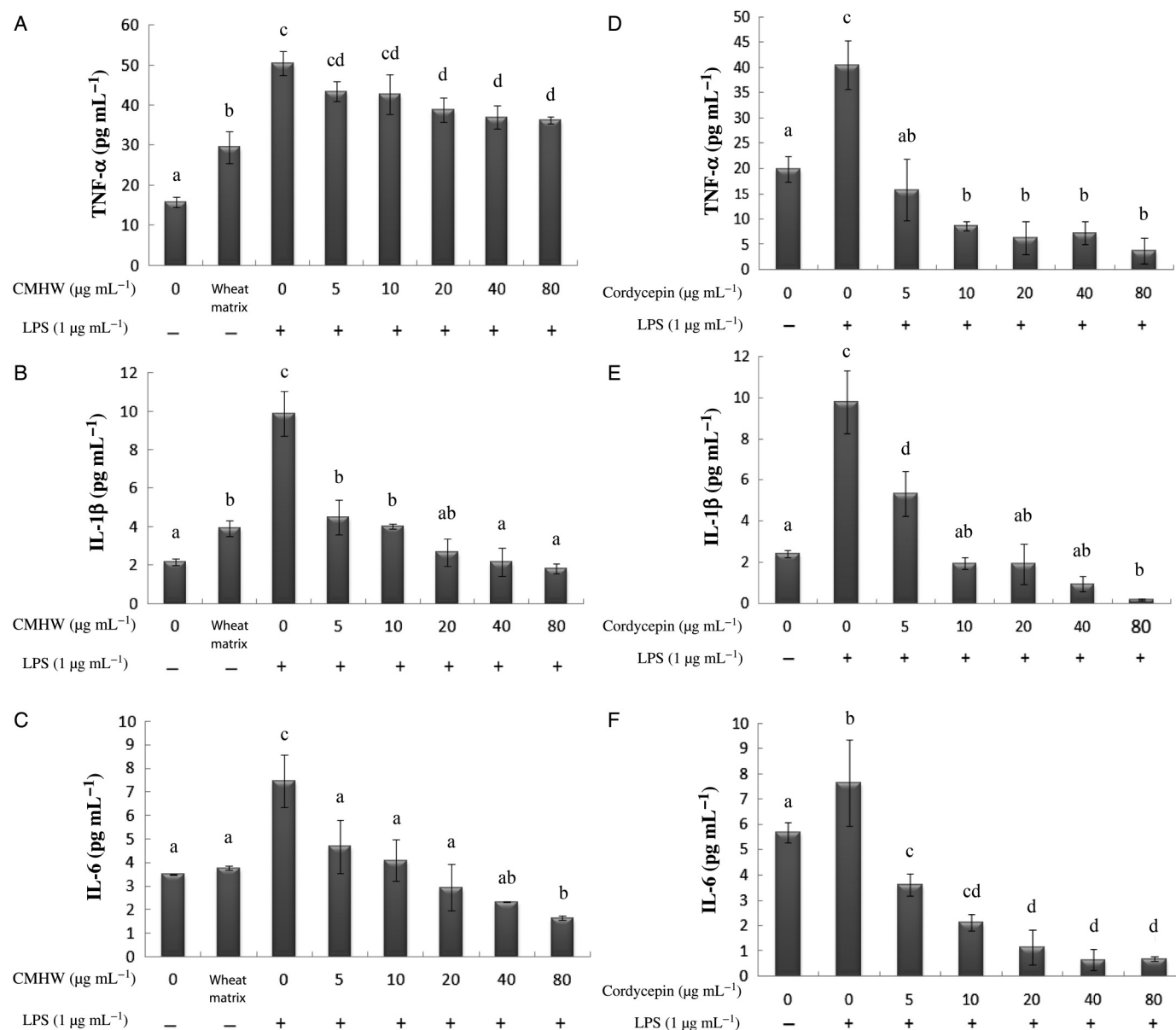
a dose-dependent manner compared with CMHW (Fig. 5B). Taken together, these results indicated that CMHW or cordycepin could inhibit LPS-induced inflammatory mediator production in PAM through suppression of the p38 MAPK signaling pathway.

Discussion

In this study, we demonstrated that CMHW could inhibit LPS-stimulated NO production and COX-2 protein expression levels in PAM. The LPS-induced phosphorylation of p38 was attenuated by CMHW, leading to reduced secretion of pro-inflammatory cytokines. The effects of CMHW on immunomodulation in PAM are similar to cordycepin.

Recent studies have demonstrated that cordycepin, the bioactive component isolated from *C. militaris* has various pharmacological activities (Sugar and McCaffrey 1998; Ahn et al. 2000; Nakamura et al. 2006). Cordycepin has been shown to have anti-inflammatory activity in LPS-stimulated Raw 264.7 macrophage cells (H.G. Kim et al. 2006; Choi et al. 2014; Zhang et al. 2014). Furthermore, it has been demonstrated that cordycepin has an inhibitory effect on pro-inflammatory cytokine expression and secretion in LPS-stimulated murine microglia (Jeong et al. 2010). Similar to previous studies, we also found that cordycepin from CMHW also inhibited pro-inflammatory cytokine secretion in LPS-stimulated PAM. Several possible molecular mechanisms for cordycepin involvement in inhibition of the

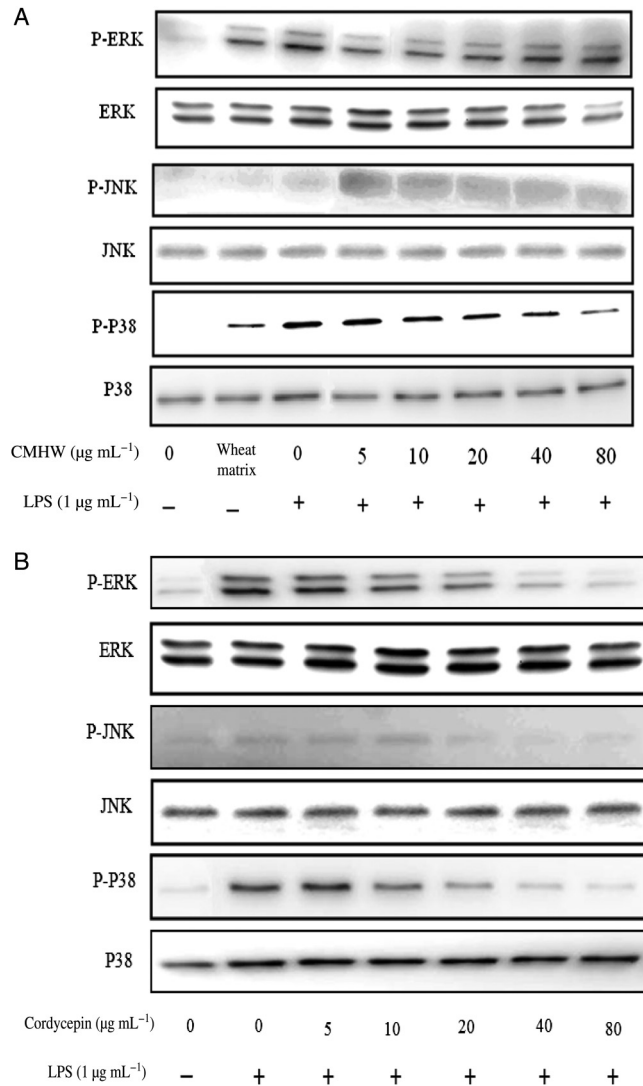
Fig. 4. Effect of *Cordyceps militaris* hot water (CMHW) and cordycepin on lipopolysaccharide (LPS) induced tumor necrosis factor- α (TNF- α) (A and D), interleukin-1 β (IL-1 β) (B and E), and interleukin-6 (IL-6) (C and F) in porcine alveolar macrophages (PAM). The PAM were pretreated with various concentrations of CMHW (0–80 $\mu\text{g mL}^{-1}$) and cordycepin (0–80 $\mu\text{g mL}^{-1}$) for 1 h and then stimulated with 1 $\mu\text{g mL}^{-1}$ LPS for 24 h. The pro-inflammatory cytokine levels in culture medium were assayed by ELISA. Values were expressed as mean $\pm$ SEM ($n = 3$). Means with different letters are significantly different ($P < 0.05$).



LPS-induced inflammatory response have been demonstrated. It has been shown that cordycepin suppresses the production of NO by attenuation of inducible NO synthase (iNOS) and COX-2 gene expression via the inhibition of phosphorylation of NF- κ B (nuclear factor kappa-light-chain-enhancer of activated B cells) activation, protein kinase B, and p38 (H.G. Kim et al. 2006; Jeong et al. 2010). The inhibitory effects of cordycepin on LPS-stimulated inflammatory responses in murine macrophages are associated with suppression of the MAPK by inhibition of the Toll-like receptor 4 signaling pathway (Choi et al. 2014). Cordycepin inhibits TNF- α production in LPS-stimulated

murine macrophage cells through activation of the adenosine monophosphate-activated protein kinase (AMPK) mediated signaling pathway (Zhang et al. 2014). It has been shown that p38 MAPK plays a key role in LPS-stimulated signal transduction pathways in macrophages (Bode et al. 2012). Here, we also demonstrate that LPS-induced phosphorylation of p38 was impaired in PAM by cordycepin from CMHW, thereby decreasing pro-inflammatory cytokine gene expression. Cumulatively, these findings demonstrate that cordycepin from CMHW alleviates production of pro-inflammatory mediators and cytokines at the transcriptional level by suppression of upstream

Fig. 5. Effect of *Cordyceps militaris* hot water (CMHW) and cordycepin on lipopolysaccharide (LPS) induced phosphorylation of mitogen-activated protein kinase (MAPK) pathway in porcine alveolar macrophages (PAM). The PAM were pretreated with (A) various concentrations of CMHW (0–80 $\mu\text{g mL}^{-1}$) and (B) cordycepin (0–80 $\mu\text{g mL}^{-1}$) for 1 h and then stimulated with 1 $\mu\text{g mL}^{-1}$ LPS for 24 h. The phosphor- or total-protein of MAPK [extracellular signal-regulated kinases (ERK), c-Jun N-terminal kinases (JNK), and p38] from cell lysates was determined by phosphor-specific or specific protein antibodies. Three experiments were carried out, and one representative result is shown.



signaling cascades, such as the MAPK and AMPK signaling pathways.

LPS can induce the production of inflammatory mediators, such as NO and prostaglandin E (Milano et al. 1995). COX-2 is a critical enzyme for prostaglandin E biosynthesis during inflammation (Ricciotti and FitzGerald 2011). Several studies have shown that iNOS and COX-2 are highly expressed in macrophages in response to inflammation (Kerwin et al. 1995; Mitchell et al. 1995).

Nitric oxide produced from macrophages by iNOS and COX-2 modulation is an important regulatory molecule during inflammation (MacMicking et al. 1997; Coleman 2001). PAM play an important role as the first defense against respiratory infections. It has been reported that inflammatory cytokines, such as TNF- α and IL-1 β , are expressed for a short period in LPS-stimulated PAM (Choi et al. 2002; Islam et al. 2013). In the present study, we also found that pro-inflammatory mediators and cytokines were increased in the culture media of LPS-treated PAM. Several functional molecules have been shown to have potent immunomodulatory effects in PAM. It has been demonstrated that β -glucans significantly increase cell activity, phagocytosis, and TNF- α production in PAM (Chaung et al. 2009). Wheat β -glucans are linear polysaccharides with repeating glucose units connected by β -1,3 and β -1,4 glycosidic linkages (Havrlentová and Kraic 2006). These water soluble β -glucans could activate macrophages by dectin-1 and Toll-like receptors, leading to pro-inflammatory mediator and cytokine production such as NO, TNF- α , and IL-1 β (Abel and Czop 1992; Brown et al. 2002, 2003; Loures et al. 2014). Consistently, our results demonstrated that wheat matrix containing soluble β -glucans was able to induce NO and cytokine production per se, whereas CMHW containing cordycepin could reverse the pro-inflammatory effects caused by wheat matrix. However, we found IL-6 levels were not affected by wheat matrix in PAM. These results were consistent with the previous reports that IL-6 production was not regulated in broilers in response to β -glucans (Guo et al. 2003). The explanation of the difference may be due to T-lymphocyte activation (Jacob and Pescatore 2014). TNF- α and IL-1 β are mainly produced from type 1 helper T-lymphocytes (Th1) in response to interleukin-12 stimulation and these cytokines are involved in the cell-mediated immunity. By contrast, IL-6 is mainly produced from type 2 helper T-lymphocytes under interleukin-4 stimulation and IL-6 is involved in the humoral immune response. Thus, wheat β -glucans may primarily induce TNF- α and IL-1 β production in PAM by activation of Th1. Furthermore, we found that CMHW significantly reduced the IL-6 production in the LPS-stimulated PAM in a concentration-dependent manner. These results were consistent with the previous reports (Jo et al. 2010). In addition, we also demonstrated that the inhibitory effect of cordycepin on pro-inflammatory was greater than CMHW in PAM. Recently, our previous study showed that feed supplementation with *C. militaris* fermentation products improves growth performance and enhances cell-mediated immunity in piglets (Cheng et al. 2016). Here, we also found that CMHW has an inhibitory effect on LPS-induced inflammatory responses in PAM. These results indicate that modulation of PAM activity by CMHW serves an effective target for immunomodulation in animals.

Water extraction is commonly considered to be a reliable extraction technique that could efficiently harvest

potent bioactive molecules. This method is also widely used for isolation and characterization of bioactive compounds from plants. It has been demonstrated that water extract of *C. militaris* promotes maturation of murine bone marrow-derived dendritic cells (G.Y. Kim et al. 2006). Hot water extract from *C. militaris* fruiting bodies exerts an inhibitory effect on NO production, TNF- α , and IL-6 secretion in LPS-stimulated murine macrophage cells (Jo et al. 2010). Consistently, we also demonstrated that CMHW attenuated pro-inflammatory cytokine secretion in LPS-stimulated PAM, including TNF- α , IL-1 β , and IL-6. Overall, these findings suggest that water extraction is a feasible method for isolation of bioactive compound from *C. militaris*.

Conclusion

In this study, we provide evidence that CMHW has an inhibitory effect on LPS-induced inflammatory responses in PAM through regulation of the p38 MAPK signaling pathway. Cordycepin produced in solid-state fermentation and extracted by hot water has high potential for development as a feed additive by providing an alternative source for immunomodulation in farm animals.

Acknowledgements

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